I KOMANG BAYU KRESNA, S.Si.

Quality Control (QC/QA) Analyst | Computational Chemist | Web & Software Developer (Vibe Coding)

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PROFESSIONAL SUMMARY

ONMIPA-PT 2025 National Chemistry Finalist and Chemistry Graduate (S.Si., Universitas Udayana, **GPA 3.77 / 4.00 — Cum Laude / With Honors**). Possesses core expertise in **Quality Control (QC) and Quality Assurance (QA) laboratory analysis for food and pharmaceutical industries**, **in silico computational chemistry**, and **AI-assisted software engineering (*Vibe Coding*)** spanning full-stack web development, scientific desktop tools, and **native PlayStation 4 (PS4) to PC game porting** (github.com/krixsan). Demonstrated hands-on proficiency operating HPLC, AAS, and UV-Vis Spectrophotometry at BPOM and accredited export inspection facilities, and formally certified across **ISO/IEC 17025, ISO 22000:2018, FSSC 22000 v7.0, GMP, and HACCP**.

EDUCATION

Universitas Udayana (UNUD) — Bachelor of Science in Chemistry (S.Si.) **August 2022 – June 2026**
GPA: 3.77 / 4.00 (Cum Laude / Dengan Pujian) · 146 Credits Completed in 8 Semesters · Graduated June 30, 2026

- **Relevant Coursework (All Grade A / 4.00):** Quantitative Analytical Chemistry, Food Chemical Analysis, Separation Methods, Laboratory K3 Safety & Management, Field Work Practice (BPOM), Chemical Toxicology, Bioactivity Assays, Phytochemistry, Biomolecules & Enzymes, Forensic Chemistry, Scientific Research Methodology.

SMA Negeri 1 Sukawati — Mathematics and Natural Sciences (MIPA) **2019 – 2022**

PROFESSIONAL & LABORATORY EXPERIENCE

Laboratory Analyst Intern — Pharmaceutical, Cosmetic & Food QC/QA **March 4 – July 4, 2025**
Balai Besar POM di Denpasar (BPOM RI) · ISO/IEC 17025 Accredited Testing Laboratory

- Conducted physicochemical quality testing on **200+ pharmaceutical, cosmetic, and food samples** utilizing High-Performance Liquid Chromatography (HPLC) and UV-Vis Spectrophotometry under strict **GMP and ISO/IEC 17025 standards**.
- Maintained instrument calibration logs, daily analytical run documentation, standard calibration curves, and volumetric standard preparations across 3 instrument systems with **zero non-conformance findings**.
- Completed official BBPOM Denpasar Internship with **Predicate "BAIK" (Good)** (Official Certificate No. **HM.03.04.13A.07.25.544**) and achieved an academic internship evaluation of **Grade A (4.00 / 4.00)**.

Quality Control (QC) Analyst — Heavy Metals Trace Analysis (AAS) & Food Safety **February – March 2025**
Bali Seafood Inspection Laboratory (BASIL) · HACCP Certified Inspection Facility

- Quantified trace heavy metal contaminants (Pb, Hg, Cd) and histamine levels across **50+ export batches** of yellowfin tuna and food products for hotels, restaurants, and airline catering using Atomic Absorption Spectroscopy (AAS), ensuring 100% compliance with **European Union and US FDA regulatory limits**.
- Executed sample acid digestion, standard curve calibration, and quality control verification under **HACCP food safety protocols**.

Teaching Assistant — Organic Chemistry Laboratory I & II **August 2025 – June 2026**
Department of Chemistry, Faculty of Mathematics and Natural Sciences, Universitas Udayana

- Instructed and supervised **100+ undergraduate students** in organic synthesis, fractional distillation, recrystallization, natural product extraction, and functional group identification.
- Enforced strict laboratory K3 safety protocols, achieving **zero safety incidents** across more than 30 practical laboratory sessions over 2 semesters.
- Evaluated 200+ student laboratory reports and led pre-lab technical safety briefings and viva sessions.

CERTIFICATIONS & PROFESSIONAL TRAINING (QA/QC & FOOD SAFETY)

- **Quality Control (QC) in Food Industry** — PT Makin Ahli Indonesia (No. 2596/AP-0P6/MA/XI-2026) **November 1, 2026**
- **FSSC 22000 v7.0 (Food Safety System Certification)** — PT Makin Ahli Indonesia (No. 5639/AP-0P4/MA/X-2026) **November 1, 2026**
- **ISO 22000:2018 (Food Safety Management System)** — PT Makin Ahli Indonesia (No. 5639/AP-0P3/MA/X-2026) **November 1, 2026**
- **Quality Assurance (QA) in Food Industry** — PT Makin Ahli Indonesia (No. 2236/AP-0P5/MA/X-2026) **October 31, 2026**
- **HACCP (Hazard Analysis and Critical Control Point)** — PT Makin Ahli Indonesia (No. 5639/AP-0P2/MA/X-2026) **October 31, 2026**
- **GMP (Good Manufacturing Practices)** — PT Makin Ahli Indonesia (No. 5639/AP-0P1/MA/X-2026) **October 31, 2026**
- **Certificate of Internship Completion (Predicate BAIK / Good)** — BBPOM Denpasar (No. HM.03.04.13A.07.25.544) **July 14, 2025**

RESEARCH & SCIENTIFIC PUBLICATIONS

Undergraduate Thesis (Skripsi) — In Silico Drug Discovery (Grade A / 4.00 · 6 Credits) 2025 – 2026

"Potensi Senyawa Ekstrak Etanol Daun Cengkeh (Syzygium aromaticum L.) sebagai Inhibitor Reseptor Estrogen Alfa (ER-α) Secara In Silico"

- Performed virtual screening (molecular docking) of **44 phytochemical compounds** identified via LC-HRMS/MS in 70% ethanol clove leaf extract against the Estrogen Receptor Alpha (ER-α, PDB ID: 3ERT) breast cancer target using **AutoDock Vina**.
- Identified 3 top hit candidates significantly outperforming the native estradiol ligand (−7.88 kcal/mol): **Naringenin (−8.415 kcal/mol)**, **(−)-Caryophyllene oxide (−8.161 kcal/mol)**, and **Ellagic acid (−8.134 kcal/mol)**.
- Verified stable hydrogen bond interactions at critical catalytic residues (Glu353 and Arg394) via **PyMOL and BIOVIA Discovery Studio**, and confirmed favorable pharmacokinetics and oral toxicity safety (Class IV–V) using **ProTox 3.0**.

VIBE CODING PROJECTS: WEB DEVELOPMENT, SCIENTIFIC TOOLS & PS4 GAME PORTING

PlayStation 4 (PS4) to Native PC Game Porting — Sengoku Basara 4 Sumeragi & J-Stars Victory VS+ 2026

Technologies: C++, C# (.NET), Vulkan 1.3 API, SDL3, x86_64 Native Runtime, CMake/Ninja · GitHub: github.com/krixsan

- Engineered native x86_64 execution runtimes and **Vulkan 1.3** presentation engines via AI-assisted *vibe coding* and clean-room reverse engineering to port PS4 titles (*Basara-4-Sumeragi-Native-PC* & *J-Stars-Victory-VS-PC*) natively to Windows PC without emulator overhead, featuring unlocked 120+ FPS framerates, high-resolution supersampling, zero-latency SDL3 gamepad polling, and custom desktop launchers.

Full-Stack Web Development via Vibe Coding (3D Research Portfolio, KasirKu POS, KostMonitor, SocialFeed) 2026

Technologies: Next.js 15, React 19, TypeScript, Tailwind CSS, Laravel 13, Prisma ORM, SQLite · Live: portofolio-mank-krisna.vercel.app

- Built and deployed production web platforms: an interactive scientific research website featuring 3D WebGL molecular docking visualizations & laboratory QC dilution calculators, a 3-in-1 UMKM Point of Sale web app (**KasirKu Smart POS**) with barcode scanning & FIFO/FEFO expiry tracking, a property management ERP (**KostMonitor**), and a community social PWA (**SocialFeed**).

DockVista PRO (v2.4) & CHEMIA (v1.5) — Scientific Computational Chemistry Desktop Applications 2026

Technologies: Python, CustomTkinter GUI, Electron, React, TypeScript, AutoDock Vina, OpenBabel, Vitest (50+ Unit Test Suites)

- Developed **DockVista PRO** to automate batch molecular docking pipelines (ligand preparation, PDBQT conversion, Vina execution, and multi-sheet Excel reporting, cutting research time by **>70%**) and **CHEMIA**, a cross-platform desktop application featuring 8 interactive chemistry simulation modules and analytical solution calculators.

HONORS & LEADERSHIP EXPERIENCE

National Finalist — ONMIPA-PT 2025 (Chemistry Division) 2025

Indonesian Talent Development Center (BPTI), Ministry of Education, Culture, Research, and Technology

- Selected to represent Universitas Udayana and advanced through regional selection (LLDIKTI Region VIII) to compete at the National Finals of the National Olympiad in Mathematics and Natural Sciences for Higher Education (ONMIPA-PT) in Chemistry.

Vice Chairman II — Chemistry Student Association (HIMAKI) Universitas Udayana 2023 – 2024

Faculty of Mathematics and Natural Sciences, Universitas Udayana

- Directed executive coordination across a 15-member core board and 3 organizational divisions to deliver academic programs, scientific seminars, and student development initiatives for 200+ undergraduate chemistry students.

TECHNICAL SKILLS

- **Laboratory Instrumentation & Wet Lab Analysis:** High-Performance Liquid Chromatography (HPLC), Atomic Absorption Spectroscopy (AAS Flame/Furnace), UV-Vis Spectrophotometry, Titrimetric & Gravimetric Analysis, Acid Digestion, Solvent Extraction, Standard Solution Preparation & Calibration Curves.
- **Quality Management & Food Safety Systems:** ISO/IEC 17025 (Testing & Calibration Laboratories), ISO 22000:2018, FSSC 22000 Version 7.0, Good Manufacturing Practices (GMP), Hazard Analysis and Critical Control Point (HACCP), SSOP, Laboratory K3 Safety Management.
- **QC Data Analysis & Documentation:** Advanced Microsoft Excel (Analytical Statistics, Linear Regression Calibration Curves, Pivot Tables, Automated QC Reporting), Microsoft Word (SOPs, Certificates of Analysis / CoA, NCR/CAPA Documentation), Microsoft PowerPoint.
- **Computational Chemistry (In Silico):** AutoDock Vina, PyMOL, BIOVIA Discovery Studio, PyRx, OpenBabel, GROMACS, ProTox 3.0, SwissADME.
- **Vibe Coding, Web Development & Game Porting:** Full-Stack Web Development (React 19, Next.js 15, TypeScript, Tailwind CSS, Laravel 13, Prisma ORM, SQLite), Native PS4-to-PC Game Porting (C++, C#, Vulkan 1.3, SDL3, x86_64 Runtime — github.com/krixsan), Desktop Apps (Python, Electron, Git).
- **Languages:** Indonesian (Native / Fluent), English (Professional Working Proficiency / Scientific Literature & International Regulatory Standards).